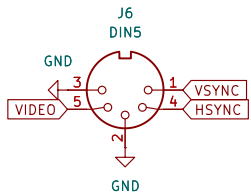
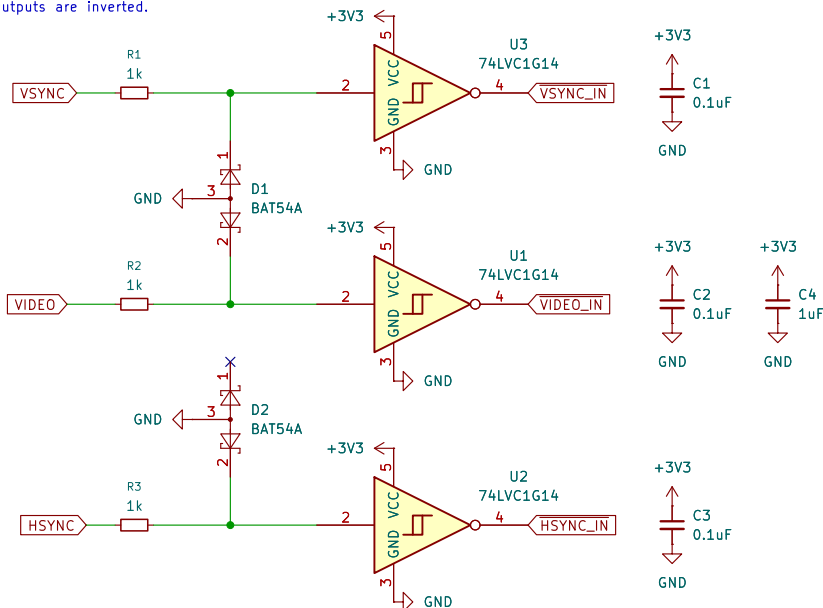


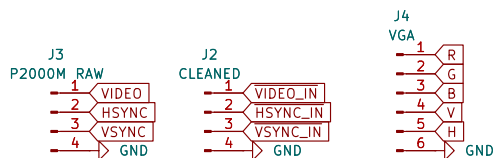
DIN5 socket. A male-to-male DIN5 cable is used to connect this board to the P2000M.



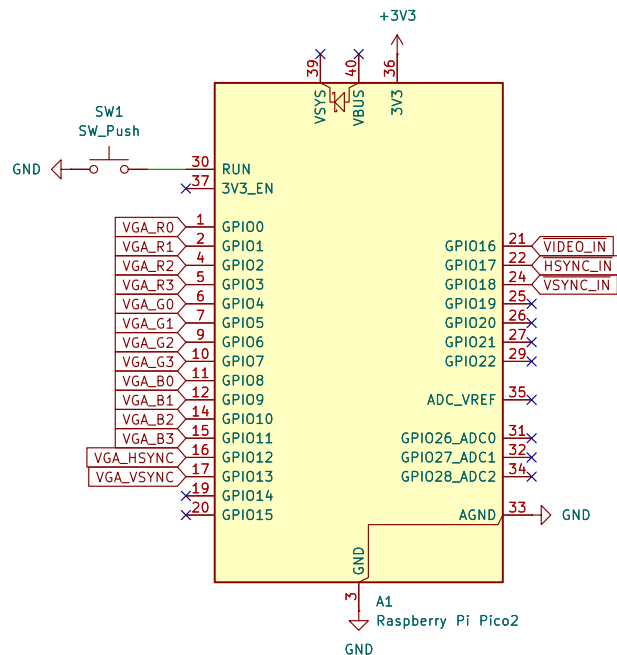
P2000M input conditioning: 1 k $\Omega$  resistors limit transient current, BAT54A diodes clamp negative undershoot, and 74LVC1G14 Schmitt-trigger inverters translate and clean the 5 V-class signals to 3.3 V RP2350 logic. Outputs are inverted.



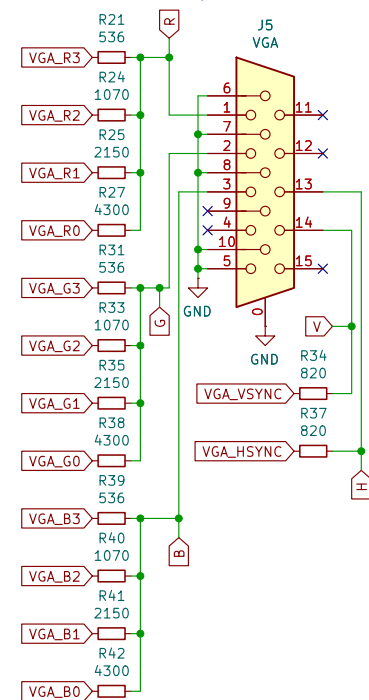
Probe pins for debugging



Pico 2 controller: GPIO16–18 capture conditioned VIDEO, HSYNC, and VSYS. GPIO0–11 provide 4-bit-per-channel RGB VGA output, with GPIO12–13 generating VGA sync. RUN has a manual reset switch; unused control, ADC, and GPIO pins are left unconnected.



VGA interface + DAC array



Sheet: /

File: p2000m-to-vga-adapter.kicad\_sch

**Title: P2000M video to VGA adapter**

Size: A4

Date: 2026-07-20

KiCad E.D.A. 10.0.3

Rev: 0

Id: 1/1